Q P Code:-25900506

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Discipline Specific Core Course (DSC) - MG3DSCMAT200 - PERSPECTIVES OF MATHEMATICS
(2024 ADMISSION ONWARDS)
Scheme For Valuation

Duration: 2 Hours **Maximum Marks: 70**

Part A
(Short Answer Questions)
Answer any 5 questions
Each question carries 2 marks

Question 1 Plot the point $(2, -\frac{\pi}{3})$ in the polar coordinates. **[A] / [CO1]**

- For correct graph ... 2 Marks

Question 2 Define a cycloid **[U] / [CO1]**

- For correct definition ... 2 Marks

Question 3 Form an equation whose roots are the negatives of the roots of the equation $x^4 - 4x^3 + 6x^2 - x + 2 = 0$ **[A] / [CO4]**

- $p(-x) = (-x)^4 - 4(-x)^3 + 6(-x)^2 - (-x) + 2$ ... 1 Mark
- Simplifying, $p(-x) = x^4 + 4x^3 + 6x^2 + x + 2$ ... 1 Mark

Question 4 Show that the equation $x^3 - x$ has 3 real roots. **[U] / [CO4]**

- Factor the left hand side: $x^3 - x = x(x^2 - 1) = x(x - 1)(x + 1)$. ... 1 Mark
- Set each factor equal to zero, we get $x = 0, x = 1, x = -1$. ... 1 Mark

Question 5 Find the area of the region R bounded by $y = x$ and $y = x^2$ in the first quadrant. **[A] / [CO5]**

- Writing the correct formula ... 1 Mark
- For finding the final answer $A = \frac{1}{6}$ after integration ... 1 Mark

Question 6 Find the equivalent integral for $\int_0^3 \int_{y^2}^9 dx\, dy$ with the order of integration reversed. **[A] / [CO5]**

- For evaluating the limits of x and y when the order is reversed ... 1 Mark
- For correct the answer $\int_0^9 \int_0^{\sqrt{x}} dy\, dx$ ... 1 Mark

Question 7 Using triple integral, find the average value of unit cube. **[A] / [CO6]**

- Any attempt ... 2 Marks

Question 8 Define Spherical coordinates. **[K] / [CO6]**

- For correct definition ... 2 Marks

Part B
(Short Essay Questions)
Answer any 5 questions
Each question carries 6 marks

Question 9 Write the polar equations in Cartesian form. Then describe or identify the graphs of both. **[A] / [CO1]**

(a) $r = \csc\theta\, e^{r\cos\theta}$

- For writing $r\sin\theta = y$ and $r\cos\theta = x$ ... 2 Marks
- For writing $r = \csc\theta\, e^{r\cos\theta} \Rightarrow r\sin\theta = e^{r\cos\theta}$ ... 1 Mark
- For writing Cartesian form $y = e^x$, the exponential curve ... 1 Mark

(b) $r\sin\left(\theta + \frac{\pi}{6}\right) = 2$

- For expanding and writing $r\sin(\theta+\alpha) = r\sin\theta\cos\alpha + r\cos\theta\sin\alpha = y\cos\alpha + x\sin\alpha$ ... 1 Mark
- For the substitution of $\cos\frac{\pi}{6} = \frac{\sqrt{3}}{2}$, $\sin\frac{\pi}{6} = \frac{1}{2}$ in $y\cos\alpha + x\sin\alpha = 2$ and reduce it to $\frac{\sqrt{3}}{2}y + \frac{1}{2}x = 2$ ... 2 Marks
- To identify $\frac{\sqrt{3}}{2}y + \frac{1}{2}x = 2$ as a straight line ... 1 Mark

(Total Marks should not exceed 6)

Question 10 Sketch the line $r\cos\left(\theta - \frac{2\pi}{3}\right) = 3$ and find the Cartesian equations for it. **[An] / [CO2]**

- Correct graph ... 2 Marks
- Expanding $\cos(\theta - \frac{2\pi}{3}) = \cos\theta\cos\frac{2\pi}{3} + \sin\theta\sin\frac{2\pi}{3}$. ... 2 Marks
- For writing $r\cos\theta\cos\frac{2\pi}{3} + r\sin\theta\sin\frac{2\pi}{3} = 2 \Rightarrow -x + \sqrt{3}y = 6$ ... 2 Marks

Question 11 If α, β, γ are the roots of the equation $x^3 + px^2 + qx + r = 0$, frame the equation whose roots are $\beta + \gamma - 2\alpha, \gamma + \alpha - 2\beta, \alpha + \beta - 2\gamma$. **[A] / [CO4]**

- Writing $\alpha + \beta + \gamma = -p$. ... 2 Marks
- Getting
  $\beta + \gamma - 2\alpha = -p - 3\alpha$
  $\gamma + \alpha - 2\beta = -p - 3\beta$
  $\alpha + \beta - 2\gamma = -p - 3\gamma$ ... 2 Marks
- Rest of the calculations to get the equation as
  $x^3 + 3(3q - p^2)x - 2p^3 + 9pq - 27r = 0$ ... 2 Marks

Question 12 Compute the possible number of real and imaginary roots of the following equations. **[A] / [CO4]**

(a) $x^4 - 8x^3 + 14x^2 + 8x - 5 = 10$

- The number of changes of sign in(μ) $f(x) = x^4 - 8x^3 + 14x^2 + 8x - 15 = 0$, is 3, The number of changes of sign in(μ') $f(-x) = x^4 + 8x^3 + 14x^2 - 8x - 15 = 0$, is 1. Hence by Descartes' Rule it has at most 4 real roots and at least 0 imaginary roots ... 3 Marks

(b) $x^5 - 2x^4 - 5x^3 + 20x^2 - 26x = 12$

- $\mu = 3, \mu' = 2$. Hence by Descartes' Rule it has at most 5 real roots and at least 0 imaginary roots ... 3 Marks

(c) $x^7 - 12x^6 + 70x^5 - 244x^4 + 519x^3 - 668x^2 + 490x = 156$

- $\mu = 7, \mu' = 0$. Hence by Descartes' Rule it has at most 7 real roots and at least 0 imaginary roots ... 3 Marks

(Total Marks should not exceed 6)

Question 13 Find $\displaystyle\int_{-2}^{0}\int_{v}^{-v} 2\,dp\,dv$. **[A] / [CO5]**

- Finding $\int_v^{-v} 2\,dp = -4v$. ... 3 Marks
- Finding $\int_{-2}^{0}(-4v)\,dv = -4\left[\frac{v^2}{2}\right]_{-2}^{0} = 8$ ... 3 Marks

Question 14 Evaluate $\displaystyle\int_0^{2\pi}\int_0^{\cos 2\theta} r^2 \sin\theta\,dr\,d\theta$. **[U] / [CO5]**

- $\displaystyle\int_0^{\cos 2\theta} r^2\,dr = \left[\frac{r^3}{3}\right]_0^{\cos 2\theta} = \frac{1}{3}(\cos 2\theta)^3$. ... 3 Marks
- Calculations upto
- $\displaystyle\int_0^{2\pi}\int_0^{\cos 2\theta} r^2 \sin\theta\,dr\,d\theta = \frac{1}{3}\int_0^{2\pi}(\cos 2\theta)^3 \sin\theta\,d\theta. = 0$ ... 3 Marks

Question 15 Evaluate the limits of integration r, θ and z for the solid enclosed by the cylinder $x^2 + y^2 = 4$, bounded above by the paraboloid $z = x^2 + y^2$ and bounded below by the xy-plane? **[A] / [CO6]**

- Conversion to cylindrical coordinates
  Cylinder: $r = 2$, Paraboloid: $z = r^2$, Plane: $z = 0$. ...3 Marks
- Determine limits of integration.
  $0 \leq \theta \leq 2\pi, \quad 0 \leq r \leq 2, \quad 0 \leq z \leq r^2$. ...3 Marks

Question 16 Sketch the region of integration and evaluate the integral $\int_0^2 \int_{x^2}^{2x} 4x + 3\, dx\, dy$. **[A] / [CO7]**

- Any Reasonable attempt ...6 Marks

Part C
(Essay Questions)
Answer any 3 questions
Each question carries 10 marks

Question 17 The ellipse $\frac{x^2}{9} + \frac{y^2}{25} = 1$ is shifted 3 units to the left and 2 units down. **[A] / [CO2]**

(a) Find the equation of the new ellipse

- For calculations to find the equation ...5 Marks
- Equation of the new ellipse $\frac{(x+3)^2}{9} + \frac{(y+2)^2}{25} = 1$. ...2 Mark2

(b) Find the vertices, foci and centre of the new ellipse

- Calculation to find the vertices at $(-3, 3)$ and $(-3, -7)$ ...3 Marks
- Calculation to find the foci at$(-3, 2)$ and $(-3, -6)$. ...2 Marks
- Find the centre at $(-3, -2)$ ...1 Mark

(c) Plot the new vertices, foci and centre and sketch in the ellipse

- Correct graph showing the vertices, foci and centre ...3 Marks

(Total Marks should not exceed 10)

Question 18 If α, β, γ are the roots of $2x^3 - 5x^2 + 4x - 1 = 0$, find: **[A] / [CO3]**

(a) $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$

- Getting
  $\alpha + \beta + \gamma = -\frac{-5}{2} = \frac{5}{2}$
  ,$\alpha\beta + \beta\gamma + \gamma\alpha = \frac{4}{2} = 2$
  $\alpha\beta\gamma = -\frac{-1}{2} = \frac{1}{2}$. ...5 Marks

- $\frac{1}{\alpha}+\frac{1}{\beta}+\frac{1}{\gamma}=\frac{\alpha\beta+\beta\gamma+\gamma\alpha}{\alpha\beta\gamma}=\frac{2}{\frac{1}{2}}=4.$ ... 3 Marks

(b) $\frac{1}{\alpha\beta}+\frac{1}{\beta\gamma}+\frac{1}{\gamma\alpha}$

- $\frac{1}{\alpha\beta}+\frac{1}{\beta\gamma}+\frac{1}{\gamma\alpha}=\frac{\alpha+\beta+\gamma}{\alpha\beta\gamma}=\frac{\frac{5}{2}}{\frac{1}{2}}=5.$ ... 3 Marks

(c) Verify that (a) $\times(\alpha\beta\gamma)$ = (b) $\times(\alpha\beta+\beta\gamma+\gamma\alpha)$

- (a) $\times(\alpha\beta\gamma)=2$ ... 2 Marks
- (b) $\times(\alpha\beta+\beta\gamma+\gamma\alpha)$=10 ... 2 Marks

(Total Marks should not exceed 10)

Question 19 (a) Form a polynomial equation with real coefficients having 1, 3 and $1+i2$ as roots. **[U] / [CO3]**

- Since the coefficients are real, the conjugate $1-2i$ is also a root. ... 2 Marks
- $f(x)=(x-1)(x-3)\big((x-1)^2+4\big)$ ... 2 Marks
- Expanding $f(x)=x^4-6x^3+16x^2-26x+15=0.$ ... 1 Mark

(b) Solve $x^3-x^2-8x+12=0$ given that it has a double root.

- Correct solution($x=2,2,-3$) by substitution or any other method ... 5 Marks

Question 20 (a) Using polar integration, find the volume of the solid region bounded above by the paraboloid $z=9-x^2-y^2$ and below by the unit circle in the xy-plane. **[A] / [CO5]**

- $V=\iint_D (9-r^2)\,dA=\int_{\theta=0}^{2\pi}\int_{r=0}^{1}(9-r^2)\,r\,dr\,d\theta.$ ... 4 Marks
- $\int_0^1 (9r-r^3)\,dr=\frac{17}{4}.$ ... 2 Marks
- $V=\int_0^{2\pi}\frac{17}{4}\,d\theta=\frac{17}{4}\cdot 2\pi=\frac{17}{2}\pi.$ ... 1 Mark

(b) Express the double integral $\int_0^1\int_0^{\sqrt{1-x^2}}(x^2+y^2)dydx$ in polar coordinates and then evaluate it.

- Getting $\int_0^1\int_0^{\sqrt{1-x^2}}(x^2+y^2)\,dy\,dx=\int_{\theta=0}^{\pi/2}\int_{r=0}^{1}r^2\,(r\,dr)\,d\theta=\int_0^{\pi/2}\int_0^1 r^3\,dr\,d\theta.$ ... 4 Marks
- Finding $\int_0^1 r^3\,dr=\left.\frac{r^4}{4}\right|_0^1=\frac{1}{4}$ ... 2 Marks
- Finding $\int_0^{\pi/2}\frac{1}{4}\,d\theta=\frac{1}{4}\cdot\frac{\pi}{2}=\frac{\pi}{8}.$ ... 1 Mark

(Total Marks should not exceed 10)

Question 21 Evaluate **[A] / [CO6]**

(a) $\displaystyle\int_0^{\sqrt{2}}\int_0^{3y}\int_{x^2+3y^2}^{8-x^2-y^2}(2x-y-z)dzdxdy$

- Getting $\displaystyle\int_{x^2+3y^2}^{8-x^2-y^2}(2x-y-z)\,dz = (2x-y)\Big[(8-x^2-y^2)-(x^2+3y^2)\Big] - \frac{1}{2}\Big[(8-x^2-y^2)^2-(x^2+3y^2)^2\Big]$. ... 3 Marks
- Rest of the calculations and the integral value $40-\dfrac{188}{5}\sqrt{2}$ ... 2 Marks

(b)
$$\int_0^{\pi/4}\int_0^{\ln(\sec y)}\int_{-\infty}^{\pi}\cos(x+y+z)dxdydz$$

- Any Reasonable attempt ... 5 Marks

Question 22 (a) Write the Jacobian of the transformation from Cartesian coordinate to cylindrical coordinate.

[A] / [CO7]

- Writing $x = r\cos\theta, \quad y = r\sin\theta, \quad z = z.$ ... 2 Marks
- $$J = \begin{vmatrix} \dfrac{\partial x}{\partial r} & \dfrac{\partial x}{\partial \theta} & \dfrac{\partial x}{\partial z} \\ \dfrac{\partial y}{\partial r} & \dfrac{\partial y}{\partial \theta} & \dfrac{\partial y}{\partial z} \\ \dfrac{\partial z}{\partial r} & \dfrac{\partial z}{\partial \theta} & \dfrac{\partial z}{\partial z} \end{vmatrix} = \begin{vmatrix} \cos\theta & -r\sin\theta & 0 \\ \sin\theta & r\cos\theta & 0 \\ 0 & 0 & 1 \end{vmatrix}.$$
  ... 2 Marks
- $J = r(\cos^2\theta + \sin^2\theta) = r.$ ... 1 Mark

(b) Find the volume of the ellipsoid

$$\frac{x^2}{2}+\frac{y^2}{2}+\frac{z^2}{2}=1$$

using an appropriate substitution.

- Any Reasonable attempt ... 5 Marks

N.B: Suitable stage marks may be fixed by the examiner for alternate correct answers.